



The scenario

A telecom company ("Telco") sells a base mobile plan and lets customers bolt on optional add-on services phone service, multiple lines, fiber vs. DSL internet, online security, online backup, device protection, tech support, streaming TV, and streaming movies. Each add-on nudges the monthly bill upward, **but not by the same amount** a fiber-internet upgrade costs far more than device protection.

The billing and revenue team wants to answer a practical question:

Given *which* add-on services a customer subscribes to, what monthly charge should we expect them to pay?

Requirements

The solution should:

1. **Load and inspect the data** read telco.csv (7,043 rows), confirm the target is target (MonthlyCharges, continuous), and verify the service columns are clean 0/1 indicators with no missing values.
2. **Define the variables** set the target as MonthlyCharges and the predictors as the ten service indicators (PhoneService, MultipleLines, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, InternetService_Fiber optic, InternetService_No).
3. **Split the data** hold out a test set (80/20) with a fixed random seed so results are reproducible.
4. **Fit a multivariable linear regression** train the model $\hat{y} = w_0 + \sum_i w_i x_i$ on the training set.
5. **Report the parameters** output the intercept (base-plan charge) and all ten coefficients (dollar contribution of each service).
6. **Evaluate on the test set** report at least R^2 and mean absolute error (MAE), and ideally RMSE.
7. **Compare to a baseline** show the multivariable model beats the original single-variable model (count of add-ons) on the same split.
8. **Interpret the results** translate coefficients into plain business terms (each weight $\approx$ the monthly price of that service).
9. **Make a prediction** given a new customer's chosen bundle, output an expected monthly charge.

Practical questions the model answers

1. **What does the base plan cost** before any add-ons? (the intercept)
2. **How much does each individual add-on contribute** to the monthly bill?
3. **Which add-on is the most expensive, and which is the cheapest?**

4. **If a customer adds fiber internet plus both streaming services**, how much should their bill rise?
5. **Given a specific bundle** (say, phone service , multiple lines, and tech support), **what monthly charge should we expect?**
6. **Is knowing *which* services a customer has meaningfully better** than just knowing *how many* they have, for predicting the bill?
7. **How accurate are our predictions** on customers the model hasn't seen within a few dollars, or off by more?
8. **Do the model's learned prices match Telco's actual price list**, and are any coefficients surprising (negative, or larger/smaller than expected)?

Deliverables

- Your work is delivered in two places: a **GitHub repository** (all the technical work) and a **zipped submission to Canvas**.

GitHub repository must contain:

- **The cleaning notebook** (.ipynb) that performs every step, with each of the 21 techniques clearly labelled by its number.
- **The cleaned dataset**, exported to clean_online_retail.csv, plus a short data dictionary of your final columns.
- **A findings summary** one page answering the seven business questions, with at least two charts.
- **Cleaning decisions log** a short list of the judgment calls you made (what you dropped, kept, or changed) and why.
- **A README.md** explaining what the project does, how to download the dataset from Kaggle, how to run the notebook, and a brief summary of your findings.
- **Requirements.txt** listing the Python libraries (and versions) needed to reproduce your work.

Submit to Canvas:

- The **link to your GitHub repository**, and
- A **single zipped folder** of the repository.